

Outbreak: Riverton

Scientific Response Director • Riverton

15 STAGES • 61 STOPS • GRADE 12 • OUTBREAK_RIVERTON

THE OPENING CARD

Three hospitals in one river city reported the same unusual illness inside a day of each other, and the ambulances are still arriving. Seven patients, four days, and the same four symptoms in every one of them. You are the scientific response director, which means what gets measured is your decision and the city acts on whatever comes back. Sofia Morales will not yet say whether seven is a lot, because nobody has written down what a normal week in this city looks like. An outbreak is only stoppable while it is still small, and this one has been running for four days.

The First Cluster

BEFORE THE STAGE — GREET

Six leads, three shifts, and a fence between them

The clinical side and the laboratory side have barely met, and by the end of the week a decision will rest on one trusting the other's number. Get round the campus and put a name to every lead before the next shift change, because at three in the morning a phone call from somebody you have met is a different conversation.

PLAN CARD 102W

Day one. Seven patients, three hospitals, four days, and the same four symptoms in every one of them. The epidemiology lead, Sofia Morales, has the surveillance printout and a reporter downstairs already has the number. This stage you decide whether this is an outbreak at all, agree what counts as a case, and spend the first of the response money. Seven sounds like a lot, and Morales will not say whether it is until she knows what a normal week in this city looks like. Call it nothing and you lose the weeks when an outbreak is still small enough to stop.

BRIEFING 34W

Three Riverton hospitals report an unusual combination of fever, dry cough, severe headache, and profound fatigue. The player must decide whether the pattern is a true outbreak signal before scarce emergency resources are committed.

WHAT YOU DECIDE 14W

Open an investigation using a defensible provisional case definition and an explicit comparison baseline.

1.1 Signal or noise?

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM ·
DIAGNOSIS

SCENE 41W

Seven patients across three hospitals, four days, and the same four symptoms. Dr. Sofia Morales, the epidemiology lead, has the surveillance printout, reporting routes and catchment figures on the desk. The press office has asked twice whether it should say anything.

GUIDE 65W

Six readings, and the quiet ones decide it. A count is not a signal until something written down beforehand says what ordinary looks like. Ask of each candidate how many of the six it covers. Two of the seven were found on manual chart review, outside the alert rule. Had every case come through that one rule, the cluster could have belonged to the rule.

ASK 11W

Is this a signal, and what on the desk decides it?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

a rate is a count measured against a population and a length of time · surveillance only sees the patients who reached it · epidemiology is the study of how health events are distributed in populations and what factors shape them · homeostasis and feedback — taken as read

VERDICT 93W

A count is not a signal. What makes it one is a comparison against something written down beforehand, and the baseline here is about two a year — so seven in four days is a departure of a couple of hundredfold, not a busy week. The reading

that matters most is the quiet one: two of the seven were found on manual chart review, outside the alert rule. Had every case arrived through that one rule, the cluster could have belonged to the rule. Two found another way is what makes it real.

TAKEAWAY 13W

A count becomes a signal only when something says what was expected instead.

1.2 Build the provisional case definition

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON · SEQUENCE

SCENE 38W

3 hospitals begin counting at 6 a.m., and each has drafted a different case definition. Those first tallies will shape the city's outbreak curve. Dr. Lena Ortiz wants one definition on every desk before the night shift leaves.

GUIDE 62W

All four steps will happen, so ask what each one lets you check. A case definition is an instrument, and it is wrong in a known direction. Too narrow and the spread is invisible. Too wide and the count fills with ordinary autumn illness. So it gets tested against known cases and against obvious non-cases, and those two tests catch opposite errors.

ASK 16W

Order the build so each step can be checked before the next one depends on it.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a curve is only meaningful if every hospital counted the same thing · a test can be wrong by missing cases or by counting the wrong ones · populations, transmission chains and exponential growth — taken as read · rates, ratios and reading data — taken as read

VERDICT 88W

A case definition is an instrument, and like any instrument it is wrong in a known direction. Too narrow and the spread is invisible; too wide and the count fills with the ordinary respiratory illness of a city in autumn. That is why it is tested twice before it is issued — against patients already known to be cases, which catches a definition that is too narrow, and against obvious non-cases, which catches one that is too wide. A definition nobody tested is a curve nobody can interpret.

TAKEAWAY 8W

Operational definitions convert vague concern into reproducible observation.

1.3 Spend the first response reserve

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON · VALUE

SCENE 31W

Dr. Sofia Morales, the epidemiology lead, has one response slot she can start this afternoon. Four proposals are ready, and tomorrow's hospital reports will arrive before any second project can begin.

GUIDE 52W

One response slot you can start this afternoon, and four proposals. Open each and ask what tomorrow's conclusions would look like with and without it. Hospital reports arrive before any second project can begin, so this purchase decides what the team can reliably say tomorrow rather than what it will eventually know.

ASK 12W

Which first purchase changes what the response team can reliably conclude tomorrow?

BEHIND THE BUTTON 122W

Why the first purchase is different. It runs while nothing else does, so it sets the baseline everything later is compared against. A measurement started tomorrow can still be useful; one that should have started today can leave a gap in the record that nothing later fills.

What 'reliably conclude' rules out. Data that arrives without a comparison — cases with no denominator, positives with no sampling frame — cannot support a conclusion however large it is. The proposals differ mostly in whether they produce a comparison or only a count.

Why the hospital reports change the calculus. They arrive whatever you buy, so a proposal that duplicates what they will contain has spent the slot on something you were getting anyway.

TAKES AS READ

three hospitals can only be combined if they count cases the same way · an independent signal can be valuable without defining an individual case

VERDICT 83W

Three hospitals are about to produce three curves. Those curves can be combined only if the same operational definition created them. A standard line list preserves dates, symptoms and denominators in one comparable structure. Wastewater adds an independent population signal, but it cannot define a patient case. Expert impressions have no common denominator, and a warning adds no evidence. The first scarce action should make tomorrow's observations comparable; the independent streams become more useful once there is a clinical baseline to compare with.

TAKEAWAY 14W

The best first measurement is the one that makes later evidence comparable and decision-ready.

END OF THE STAGE 20W

Use baseline variation, case definition, controls, correlation versus coincidence, measurement bias as an evidence chain rather than as isolated facts.

What Kind of Agent?

BEFORE THE STAGE — TRIAL

Walk the campus before you are needed on it

Week three, and the hospital has been rebuilt around the outbreak: triage marquees in the courtyards, a decon tunnel across the main route, container labs where the car park was. Ortiz, the clinical physiology lead, walks every arrival round once, because a route you have not walked is a route you will take through a clean corridor in dirty gloves.

PLAN CARD 105W

Day two. Pathology has four results from the same specimens and will not put a name to any of it. The microscope images are the most convincing thing anybody has seen all week, which is exactly why the cell biology lead, Maya Chen, does not trust them yet, and her diagnostics counterpart Ravi Patel wants to commit the diagnostics bench tonight. This stage you decide what broad kind of thing this is and buy the one piece of evidence that would settle it. Everything downstream is chosen off this call: what to grow it in, which drugs are worth considering, what tomorrow's test looks for.

BRIEFING 33W

Initial samples contain biological material, but the microscope and the routine tests disagree. The team must distinguish a cellular pathogen from a virus or a noninfectious cause without relying on one ambiguous image.

WHAT YOU DECIDE 11W

Classify the broad agent category and identify the next discriminating test.

2.1 What kind of agent fits the whole panel?

CLINICAL PHYSIOLOGY · RIVERTON GENERAL — CLINICAL WING · A ROOM ·
DIAGNOSIS

SCENE 39W

Pathology has four results from the same positive specimens and still cannot name the agent class. The samples are waiting under the hood. Dr. Lena Ortiz needs a defensible broad classification before the diagnostic team commits its overnight run.

GUIDE 56W

Five readings, and three of them are negatives doing the classification. No cellular forms. No bacterial marker. No fungal wall. Ask of each candidate how many of the five it covers. The signal rises only with living host cells, and the extraction blank is clean. Note what this cannot reach: which nucleic acid, or which agent.

ASK 10W

Which explanation fits every reading, including the quiet negative controls?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

cells and viruses differ in whether they can copy themselves alone · a negative control is a result, not an absence of one

VERDICT 78W

The negative cellular tests do most of the classification work. No cellular forms are seen, the bacterial marker is absent, and the fungal-wall assay is negative. The signal rises only with living host cells, which supports an agent that depends on host machinery. A clean extraction blank makes laboratory carryover less likely. Together these observations support a virus-like class. They do not establish whether the genome is RNA or DNA, and they do not identify the exact agent.

TAKEAWAY 19W

Agent classification should combine cellular structure, host dependence, molecular evidence, and controls rather than rely on one striking image.

2.2 From specimen to broad classification

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON · SEQUENCE

SCENE 43W

The microscope bench has patient material, healthy tissue and a negative control on the same bench, and four things it could do with them. Ortiz wants the order written down before anybody touches a specimen, because two of these consume what they measure.

GUIDE 58W

All four steps will be done, so ask what each one needs to be true first. Two of them consume what they measure. Measure a structure in a contaminated preparation and you have measured the contamination at high magnification. And asking whether something copies itself alone only means anything once you know whether it is cellular at all.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a measurement of a contaminated preparation measures the contamination

VERDICT 69W

Each step is only interpretable if the one before it held. Comparing against controls first is what separates the agent from the specimen it arrived in — measure a structure in a contaminated preparation and you have measured the contamination, at high magnification. Size and structure then say whether this is cellular at all, and only once you know that does asking whether it copies itself alone mean anything.

TAKEAWAY 14W

A good classification is an evidence synthesis, not a label generated by one instrument.

2.3 When heat changes an enzyme assay

MOLECULAR DIAGNOSTICS · A ROOM · CHOICE

SCENE 30W

A separate reporter-enzyme assay runs normally after a 37-degree pre-incubation and loses most activity after the same enzyme is held at 55 degrees. Substrate, buffer and reading method are unchanged.

GUIDE 61W

Four options, and only one thing changed between the two runs. Ask of each option whether it names something that was actually varied. Substrate, buffer and readout are the same. The patient sample and the target sequence were not heated. So the answer has to be about the thing that was, and the two temperatures pull in opposite directions on it.

ASK 9W

What does the temperature comparison show about the catalyst?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a protein's function depends on it keeping its shape · a buffer is a chemical system that resists large pH changes when acid or base is added · cell structure and function — taken as read

VERDICT 80W

Enzyme activity depends on both molecular motion and protein structure. Modest warming can speed collisions, but enough heat can disrupt the shape that creates the active site. Here the same reporter enzyme, substrate, buffer and readout are used after two pre-incubation temperatures. The cleanest interpretation is loss of enzyme activity at the higher temperature. This is not a PCR cycling rule, and it does not

show that a target nucleic acid was destroyed. The experiment is specifically about the catalyst.

TAKEAWAY 15W

A controlled comparison can show which part of an assay is sensitive to changed conditions.

END OF THE STAGE 21W

Use cell structure, viruses versus cells, prokaryotes and eukaryotes, staining, independent evidence as an evidence chain rather than as isolated facts.

The Point of Entry

BEFORE THE STAGE — FOLLOW

Chen walks the specimen path once, end to end

A swab taken in a marquee reaches the container lab through four pairs of hands, and every handover is a place contamination can enter. Chen, the cell biology lead, walks the whole path this morning at the pace it actually happens. Stay with her — the handovers are where the walk is learned and where somebody following too closely gets in the way.

PLAN CARD 102W

Day four. The agent reaches the airway in everybody and only some cells are being damaged, and Adeyemi has spent two nights on why. This stage you work out why those cells and not their neighbours, and design the experiment that separates two steps people keep merging. Two of the three benches are already booked and Adeyemi wants the third tonight. Chen has asked for the resistant cells to be kept aside whatever is decided. Your teams start designing a therapy from this picture tonight. If the model is wrong they spend three months blocking a step the agent does not need.

BRIEFING 26W

The suspected agent reaches respiratory tissue, but only some cell types become affected. The player must determine how membrane structure, receptors, and transport explain this selectivity.

WHAT YOU DECIDE 18W

Explain why one cell type is vulnerable and select the experiment that separates receptor binding from downstream replication.

3.1 Crossing the cellular boundary

CELL & MEMBRANE BIOLOGY · A ROOM · SEQUENCE

SCENE 40W

Only some cell types in the respiratory tissue are affected, and the culture wing has four cards describing what happens when something crosses into a cell. Nobody has put them in order yet, and the therapy meeting is at four.

GUIDE 60W

These four are one physical chain, so ask what has to happen before each can. Nothing enters until something on the surface is recognised, and a cell without the right protein cannot be infected however much material arrives. That is why only some cell types are affected. Everything the cell then does wrongly is downstream of a barrier already crossed.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a cell membrane is a selective barrier, not a wall · cell structure and function — taken as read

VERDICT 70W

The order is physical, and it is why the tissue pattern is predictable. Nothing enters until something on the surface is recognised, and a cell without the right protein cannot be infected however much material reaches it. Recognition comes first, the membrane rearranges second because that is what binding triggers, material crosses third, and everything the cell then does wrongly is downstream of a barrier that has already been crossed.

TAKEAWAY 10W

Membranes are selective interfaces whose proteins can determine cell vulnerability.

3.2 Choose the membrane mechanism

CELL & MEMBRANE BIOLOGY • A PERSON • PROTOCOL

SCENE 35W

Four observations from the culture bench each describe something moving across a membrane. The cell biologist wants each one named by its mechanism before the entry model goes further, without guessing from the molecule names.

GUIDE 58W

Four observations and four mechanisms, and the name of the molecule tells you nothing. Pair them on three tests. Which way does it go, downhill or uphill? Is a protein involved, or does it cross the lipid directly? And is it too big for any channel? The ATP being consumed is the receipt for one of those answers.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a concentration gradient has a direction • ATP is the cell's energy currency • cell structure and function — taken as read

VERDICT 73W

Three constraints tell these apart and none of them is the name of the molecule. Which way it goes: downhill needs no energy, uphill does, and the ATP consumption is the receipt. Whether a protein is involved: a small nonpolar molecule crosses the lipid directly, an ion cannot and needs a pore. And how big it is: a bound particle too large for any channel is taken in by the membrane wrapping it.

TAKEAWAY 12W

Transport mechanisms are inferred from constraints, not memorized from molecule names alone.

3.3 Separate binding from entry

CELL & MEMBRANE BIOLOGY · A PERSON · CONTROL

SCENE 32W

Dr. Maya Chen, the cell biology lead, has a simulated labelled-particle entry assay on vulnerable airway cells. Three perturbations are ready, and the therapy meeting needs evidence about the first physical step.

GUIDE 66W

The number you are watching is the entry signal: how much labelled particle ends up inside the cells. Three things can be changed, and only one of them acts on the step the meeting is asking about. A response counts only if it is larger than the well-to-well scatter of the assay. Change one, run it, restore it, and name the change the entry signal follows.

ASK 17W

Which single change makes the entry signal follow the receptor interaction rather than exposure or antibody handling?

BEHIND THE BUTTON 166W

What the assay measures, and what it does not. The label reports particles that have got inside the cell, not particles stuck to the outside. That distinction is the whole point of the experiment: binding to the surface and entering the cell are separate steps, and a therapy that blocks one may do nothing about the other.

Why a matched antibody is in the list. It is there to be changed and produce nothing. An antibody of the same class that does not target the receptor controls for everything handling an antibody involves — the buffer, the incubation, the washing — so a response to the real blocker means something only if this one gives none.

Why the dose is in the list too. Give more particles and more get in, which moves the number without saying anything about the mechanism. It is the plausible confound: a change that moves the readout for a reason that has nothing to do with the step being asked about.

TAKES AS READ

a causal test changes one factor while holding the others fixed · a reversible effect is stronger evidence than a one-way coincidence · transcription, translation and gene expression — taken as read

VERDICT 83W

Receptor abundance is only an indirect clue. A causal test asks whether the proposed interaction is required for entry. Blocking that interaction while keeping dose and handling fixed sharply lowers the entry signal. A matched antibody leaves the signal unchanged, which controls for adding an antibody at all. Washing the blocker away restores entry, so the effect follows the manipulated link in both directions.

Doubling exposure changes signal size but does not explain why the receptor-specific perturbation removes entry under otherwise matched conditions.

TAKEAWAY 19W

A causal mechanism is stronger when changing one proposed link changes the response and reversing that change restores it.

3.4 Signal or noise?

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM · CALLBACK
· DIAGNOSIS

SCENE 41W

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that matters most is the quiet one: two of the seven were found on manual chart review, outside the alert rule. Had every case arrived through that one rule, the cluster could have belonged to the rule. Two found another way is what makes it real.

TAKEAWAY 13W

A count becomes a signal only when something says what was expected instead.

END OF THE STAGE 19W

Use phospholipid bilayer, receptors, diffusion, facilitated transport, endocytosis, cell specificity as an evidence chain rather than as isolated facts.

Hijacked Cells

PLAN CARD 103W

Day six. Infected cells are turning out RNA and proteins the sequencing centre does not recognise, 1 of them at 40 times the level of anything beside it. A genomic epidemiologist, Lena Bergström, has the time- course data and thinks everybody is looking at the wrong molecule; she has had it for two days and nobody has asked for it. This stage you decide where in the cell's own production line the agent is interfering. The loudest molecule in a sample is often a consequence rather than a cause. Aim at the wrong stage and the whole molecular programme goes into a bystander.

BRIEFING 33W

Molecular data suggest that affected cells are producing unfamiliar RNA and proteins. The player must reconstruct how information flows from nucleic acid to protein and identify where the agent interrupts normal cell function.

WHAT YOU DECIDE 16W

Build the information-flow model and choose a measurement that distinguishes more RNA from more efficient translation.

4.1 From genetic information to protein

MOLECULAR DIAGNOSTICS · A ROOM · SEQUENCE

SCENE 45W

Affected cells are making RNA and proteins the sequencing centre does not recognise. Four cards describe how a cell gets from a gene to something that does a job. Nobody can say where the change is until the chain is on the board in order.

GUIDE 60W

These four are one chain, and abundance can change at any stage independently. That is why the order is worth stating rather than treating expression as one event. Ask what each card needs to exist first. A template has to be reachable. A transcript has to exist. And a chain of amino acids does nothing at all until it folds.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

genes are read into RNA and RNA is read into protein · DNA structure and replication — taken as read

VERDICT 72W

Abundance can change at any one of these stages independently, which is exactly why the chain is worth stating rather than treating expression as one event. The template has to be reachable before anything transcribes it; a transcript has to exist before a ribosome can read it; and a chain of amino acids does nothing at all until it folds. Each stage is a separate place for the answer to be hiding.

TAKEAWAY 13W

Changes in protein abundance can arise at several distinct stages of gene expression.

4.2 Locate the molecular change

MOLECULAR DIAGNOSTICS · A PERSON · PROTOCOL

SCENE 33W

Four patterns from the gene-expression bench each pair one measurement with another. The cell biologist wants each pattern placed at the stage it reports on before anyone proposes a host pathway to target.

GUIDE 63W

Four patterns on the left, and four places the change could sit. Pair them by asking which two measurements each pattern compares, and which direction they disagree in. Message up with protein flat is one end. Protein up with message flat is the other. And a protein present in normal amounts but not working is a claim about the molecule, not the quantity.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

the amount of a protein and the amount of its message are measured separately ·
dNA structure and replication — taken as read

VERDICT 87W

RNA abundance, protein abundance, sequence and activity are four different measurements, and the central dogma is a chain of transformations rather than a promise that they move together. Message up with protein flat means the message is not being turned into protein, or the protein is being destroyed as fast as it is made. Protein up with message unchanged means the opposite end. And a protein present

in normal amounts but not working is a claim about the molecule, not about how much of it there is.

TAKEAWAY 17W

The central dogma is a chain of measurable transformations, not a claim that every stage changes together.

4.3 Find the controlling stage

MOLECULAR DIAGNOSTICS · A PERSON · CHOICE

SCENE 39W

A familiar human protein is fortyfold more abundant in affected cells than in matched controls. Its messenger RNA has not been quantified, the core can fund two measurements, and the team is already arguing about which host pathway changed.

GUIDE 64W

Four measurements, and the protein result is compatible with all of the mechanisms. Ask of each option where it sits in the chain: upstream of the others, or downstream. Two of these test what happens to a transcript that nobody has counted yet. The core can fund two measurements, and the first should be the one that tells you where to spend the second.

ASK 11W

Abundance alone fits every mechanism. Which measurement do you make first?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a protein's abundance is set by how fast it is made and how fast it is destroyed · dNA structure and replication — taken as read

VERDICT 84W

A fortyfold protein increase is compatible with several mechanisms: more messenger RNA, more protein made from each RNA molecule, or slower protein degradation. The first measurement should sit upstream of the other two. Quantifying messenger RNA over the same time course tests whether transcript abundance moved with the protein. If RNA also rises strongly, increased transcription becomes a leading contributor. If RNA stays flat, the investigation moves downstream to translation

efficiency and protein turnover instead of pretending the protein result already identified the cause.

TAKEAWAY 12W

Protein abundance alone cannot distinguish increased transcription, changed translation, or altered degradation.

4.4 Build the provisional case definition

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM · CALLBACK
· SEQUENCE

SCENE 38W

3 hospitals begin counting at 6 a.m., and each has drafted a different case definition. Those first tallies will shape the city's outbreak curve. Dr. Lena Ortiz wants one definition on every desk before the night shift leaves.

GUIDE 62W

All four steps will happen, so ask what each one lets you check. A case definition is an instrument, and it is wrong in a known direction. Too narrow and the spread is invisible. Too wide and the count fills with ordinary autumn illness. So it gets tested against known cases and against obvious non-cases, and those two tests catch opposite errors.

ASK 16W

Order the build so each step can be checked before the next one depends on it.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a curve is only meaningful if every hospital counted the same thing · a test can be wrong by missing cases or by counting the wrong ones · populations, transmission chains and exponential growth — taken as read · rates, ratios and reading data — taken as read

VERDICT 88W

A case definition is an instrument, and like any instrument it is wrong in a known direction. Too narrow and the spread is invisible; too wide and the count fills with the ordinary respiratory illness of a city in autumn. That is why it is tested twice before it is issued — against patients already known to be cases, which catches a definition that is too narrow, and against obvious non-cases, which catches one that is too wide. A definition nobody tested is a curve nobody can interpret.

TAKEAWAY 8W

Operational definitions convert vague concern into reproducible observation.

END OF THE STAGE 20W

Use DNA and RNA, transcription, translation, codons, protein function, gene regulation as an evidence chain rather than as isolated facts.

A Test Before Morning

PLAN CARD 90W

Day nine, and the hospitals need a working test by sunrise. Patel's overnight run has flagged several patients positive — and one of the controls, a tube that had nothing in it and should have come back empty. 3000 people have already been told to isolate on that run and the bulletin goes out at 6. This stage you decide whether the run can be believed and what a positive would mean for a patient. Patel wants to re-run everything from the start. The hospitals want the six o'clock number.

BRIEFING 26W

Hospitals need a rapid diagnostic by sunrise. The molecular signal is promising, but contamination, false negatives, and prevalence could make a technically impressive test clinically misleading.

WHAT YOU DECIDE 13W

Approve a diagnostic workflow only after its analytical and clinical limitations are explicit.

5.1 Build a trustworthy amplification run

MOLECULAR DIAGNOSTICS · A ROOM · SEQUENCE

SCENE 43W

Hospitals want a diagnostic by sunrise and the sample-processing room has four hours. Four steps are on the whiteboard. Amplification multiplies whatever is in the tube by a factor of 1 billion, which is the reason this order is argued about at all.

GUIDE 56W

All four steps will happen, so ask what each one protects. The amplification multiplies whatever is in the tube, including whatever came from the room. Identity has to survive extraction or the result belongs to nobody. And controls have to be in the run rather than added afterwards, which is what makes this an ordering question.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

amplification copies whatever template is present, wanted or not · DNA structure and replication — taken as read

VERDICT 79W

A patient result is the output of a controlled process, and read before the controls are read it is an instrument reading rather than evidence. Identity has to survive extraction or the result belongs to no one. The controls have to be in the run, not added afterwards — a positive control shows the chemistry worked, and a no-template control shows the signal did not come from the room. Only when they behave is the patient well worth interpreting.

TAKEAWAY 15W

A diagnostic result is the output of a controlled process, not simply a machine reading.

5.2 Why did the assay turn positive?

MOLECULAR DIAGNOSTICS · A PERSON · TRACE

SCENE 36W

Dr. Ravi Patel has patient wells, two controls and an independent platform on one screen. One ward has already cleared isolation rooms. If this run is wrong, patients will be moved before the morning team arrives.

GUIDE 60W

Open each result to see what its run depended on. Keep the ones that would still stand if one workflow step were contaminated, and untick the rest. Then name the dependency the failing ones share. A ward is already clearing isolation rooms, so both halves matter: a false positive moves patients, and discarding a true one leaves an outbreak unwatched.

ASK 12W

Which results share one workflow dependency, and what independent evidence survives it?

BEHIND THE BUTTON 137W

How a shared workflow makes results agree wrongly. Wells processed in the same batch pass through the same extraction, the same reagents and the same instrument. Contamination anywhere in that chain appears in every well downstream of it, and the pattern looks like a cluster of genuine positives.

What the controls are actually for. A negative control that stays negative rules out contamination affecting the whole batch. A positive control shows the chemistry worked. Neither can rule out a contamination event that entered after the controls were placed, which is why the independent platform is on the screen.

Why independence beats repetition. Running the same assay again on the same extract shares every dependency, so it will reproduce a contaminated result perfectly. A different platform on a different extract is the only evidence that survives the question.

TAKES AS READ

a no-template control contains no patient sample · two methods with different chemistry can fail for different reasons · DNA structure and replication — taken as read

VERDICT 86W

A no-template control has no patient material, so a target signal there must come from the workflow or reagents. That same path is shared by the patient amplification wells, which makes their positive signals ambiguous. The known positive control still shows that the chemistry can amplify target, but it cannot rescue a contaminated run. The independent platform uses different preparation and chemistry, so its negative result remains informative. The correct inference comes from tracing dependencies, not from counting how many wells on one plate are positive.

TAKEAWAY 13W

Agreement is not independent evidence when several results inherit the same contamination path.

5.3 What does a positive mean?

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM · BALLPARK

SCENE 30W

The mayor's office wants to screen a low-prevalence population and treat every positive as a case. The statistics desk has a ten-thousand-person validation table with reference status and assay results.

GUIDE 60W

Six numbers, and two of them are tempting figures from other situations. Ask of each whether this table supplies it. Sensitivity and specificity are properties of the test. What a positive is worth is not, because it depends on how many people had it to begin with. That is why a low-prevalence screen produces more false positives than true ones.

ASK 11W

Estimate the chance that a positive result is a true case.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

prevalence is the share of a population that has the condition · sensitivity and specificity are properties of a test, not of a population

VERDICT 73W

The validation table lets all three quantities be computed. Among 100 infected people, 90 test positive and 10 test negative, giving 90% sensitivity. Among 9,900 uninfected people, 9,801 test negative and 99 test positive, giving 99% specificity. The same counts also show why a positive is less decisive in a low-prevalence group: there are 90 true positives but 99 false positives. So only 90 of 189 positive tests are true cases, about 48%.

TAKEAWAY 12W

Positive predictive value depends on prevalence as well as sensitivity and specificity.

5.4 What the marquee is taking now

CLINICAL PHYSIOLOGY · RIVERTON GENERAL — CLINICAL WING · A ROOM · SPOT

SCENE 42W

Arrivals come through the marquee all afternoon. The triage rule is on a whiteboard by the entrance, and the clinical lead rewrites it as the ward state changes — after the beds fill, after the swab results land, after the second ambulance.

GUIDE 58W

Take the arrivals the current rule wants and leave the rest. The rule on the whiteboard is replaced during the afternoon and nobody calls it out. What is scored is the arrivals either side of a change, because they are the only ones that show whether the marquee is reading the board or working from the last hour.

ASK 13W

Sort arrivals to the rule on the whiteboard, and keep watching the whiteboard.

BEHIND THE BUTTON 101W

Why a triage rule changes at all. Early in a shift the rule sorts by how ill somebody is. Once isolation beds run out it sorts by who can be safely sent home to wait. After the swabs come back it sorts by who is confirmed, because a confirmed case and a suspected one no longer belong in the same queue.

Why the changeover is where harm happens. A marquee working to the old rule for ten more minutes routes confirmed cases into the queue the new rule was written to empty, and nothing on the board records that it happened.

TAKES AS READ

a triage rule can be rewritten while patients are still arriving

VERDICT 127W

Three rules run across the afternoon and each wants a different part of the board. Anybody short of breath, then anybody who cannot isolate at home, then anybody with a confirmed swab. A patient can answer two of those at once, which is what makes the switch cost real rather than notional. What the panel scores is the window either side of each change, because most arrivals are wanted by neither rule and are handled correctly by somebody who has read nothing. The marquee's own version of this is the quarter of an hour after the isolation beds fill: the rule has changed, and the people already being walked down the corridor are being walked there under the rule that was in force when somebody set off.

TAKEAWAY 15W

The cost of a withdrawn instruction is paid by whoever is still working to it.

5.5 Crossing the cellular boundary

CELL & MEMBRANE BIOLOGY · A ROOM · CALLBACK · SEQUENCE

SCENE 40W

Only some cell types in the respiratory tissue are affected, and the culture wing has four cards describing what happens when something crosses into a cell. Nobody has put them in order yet, and the therapy meeting is at four.

GUIDE 60W

These four are one physical chain, so ask what has to happen before each can. Nothing enters until something on the surface is recognised, and a cell without the right protein cannot be infected however much material arrives. That is why only some cell types are affected. Everything the cell then does wrongly is downstream of a barrier already crossed.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a cell membrane is a selective barrier, not a wall · cell structure and function — taken as read

VERDICT 70W

The order is physical, and it is why the tissue pattern is predictable. Nothing enters until something on the surface is recognised, and a cell without the right protein cannot be infected however much material reaches it. Recognition comes first, the membrane rearranges second because that is what binding triggers, material crosses third, and everything the cell then does wrongly is downstream of a barrier that has already been crossed.

TAKEAWAY 10W

Membranes are selective interfaces whose proteins can determine cell vulnerability.

END OF THE STAGE 18W

Use PCR logic, controls, sensitivity, specificity, predictive value, prevalence as an evidence chain rather than as isolated facts.

Why Are Some Patients Sicker?

BEFORE THE STAGE — HUNT

Six ventilators, and the board accounts for four

Before tonight's admissions the ward has to know what it actually has. Four ventilators are where the board says. Two were moved during the weekend surge and nobody updated anything. Find them, because a machine nobody can locate is a machine that does not exist at two in the morning.

PLAN CARD 102W

Day 13. Nakamura has two patients who arrived with the same complaint and the same test results. One is walking to the bathroom. The other is failing, and their oxygen saturation reads normal. This stage you find which link in the oxygen chain has broken and buy the measurement that proves it. Nakamura has one arterial line and two patients she would like it in. The second patient's heart rate has not moved all morning, which she finds harder to explain than the saturation. Escalate the wrong link and the treatment is aimed at a part of the chain that is working.

BRIEFING 25W

Patients with similar test results have radically different oxygen levels and recovery times. The team must connect cellular damage to organ-system physiology and homeostatic failure.

WHAT YOU DECIDE 16W

Identify the physiological bottleneck and select measurements that distinguish ventilation, diffusion, circulation, and cellular oxygen use.

6.1 Locate the oxygen-delivery failure

CLINICAL PHYSIOLOGY · RIVERTON GENERAL — CLINICAL WING · A ROOM · CHAIN

SCENE 39W

A patient's oxygenation is worsening while Dr. Lena Ortiz reviews the bedside console. It shows PaCO₂ 39 mmHg, low arterial oxygen, haemoglobin 14 g/dL, cardiac output 5.2 L/min and new lung imaging. The team needs the bottleneck before rounds.

GUIDE 61W

Build the path oxygen takes, in order, from the air in the room to the tissue that uses it. Then name the link where this patient's chain fails. Read the panel rather than the diagnosis: the haemoglobin, the cardiac output and the carbon dioxide are each telling you that a particular link is working, and only one of them is not.

ASK 11W

Where does the oxygen-delivery chain most likely fail in this panel?

BEHIND THE BUTTON 157W

What the chain is. Air reaches the alveoli by ventilation, oxygen crosses into the blood by diffusion, haemoglobin carries it, the heart delivers it, and the tissue takes it up. Each step can fail on its own, and the treatment for each is different — which is why 'low oxygen' is a symptom rather than a diagnosis.

How to read the panel as evidence about links. A normal carbon dioxide says the patient is moving air, so ventilation is not the failure. Haemoglobin of 14 says the carrying capacity is there. A cardiac output of 5.2 litres a minute says delivery is working. Each normal value eliminates a link.

Why the imaging matters most here. New changes in the lungs point at the step between the alveolus and the blood. That is the one link the other measurements cannot exonerate, and it is the one whose treatment differs most from what would be given for the others.

TAKES AS READ

oxygen must move from air to alveoli, blood and tissues in sequence · a normal reading at one link makes a simple failure there less likely · homeostasis and feedback — taken as read

VERDICT 86W

Oxygen delivery is a chain, so no single normal value proves the patient is safe. Near-normal PaCO₂ makes isolated global hypoventilation less likely. Haemoglobin of 14 g/dL is adequate for carriage, and a cardiac output of 5.2 L/min does not support a simple low-flow explanation. The abnormal evidence sits at the lung handoff: arterial oxygen is low and imaging shows diffuse alveolar involvement. Mixed disorders remain possible, so the next step is to measure downstream consequences rather than treat one normal link as a complete clearance.

TAKEAWAY 20W

A transport chain is localized by tracing every required handoff and using normal links to narrow where the bottleneck lies.

6.2 Estimate oxygen delivery

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM · BALLPARK

SCENE 33W

The bedside argument has stalled on which of two patients is worse off. Both have the same oxygen content per litre of blood; one moves three litres a minute and the other five.

GUIDE 54W

Five numbers, and three of them are the other patient's flow, the saturation and the haemoglobin. Ask of each whether this delivery depends on it. Both patients carry the same oxygen per litre, so the concentration cannot separate them. A concentration is not a rate, and the flow term is doing all the work.

ASK 11W

Estimate oxygen delivery for the patient with the lower cardiac output.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a rate is an amount multiplied by how fast it is delivered · homeostasis and feedback — taken as read

VERDICT 28W

Two patients with identical saturations can differ by a factor of two in delivery, because the flow term is doing the work. A concentration is not a rate.

TAKEAWAY 12W

A normal concentration does not guarantee a normal delivery rate; flow matters.

6.3 Read the downstream acid–base signal

CLINICAL PHYSIOLOGY • RIVERTON GENERAL — CLINICAL WING • A PERSON • BALLPARK

SCENE 30W

The oxygen-delivery panel points toward a lung bottleneck, but the team still needs a downstream consequence. Arterial bicarbonate is 14 mmol/L and PaCO₂ is 31 mmHg; lactate is still pending.

GUIDE 58W

Six numbers, and two of them are normal values rather than this patient's: a middle-of-range pH, and a normal carbon dioxide. Ask of each whether the relationship uses it. The dissolved gas is the carbon dioxide times its solubility, so those two go together. And note what the answer settles: a disturbance and a compensation, not a cause.

ASK 11W

Estimate the pH from bicarbonate 14 mmol/L and PaCO₂ 31 mmHg.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

bicarbonate is the major measured base in this buffer pair • PaCO₂ is the arterial carbon-dioxide pressure in mmHg

VERDICT 78W

For the blood bicarbonate buffer, dissolved carbon dioxide is proportional to PaCO₂. The useful clinical form is $\text{pH} = 6.1 + \log_{10}([\text{HCO}_3^-]/(0.03 \times \text{PaCO}_2))$. With bicarbonate 14 and PaCO₂ 31, the ratio is about 15 and the pH is about 7.28. The low bicarbonate supports a metabolic acid–base disturbance, while the reduced PaCO₂ moves pH upward as respiratory compensation. The calculation does not identify the cause by itself; the pending lactate and the rest of the oxygen-delivery evidence still matter.

TAKEAWAY 16W

Acid–base status reflects both bicarbonate and carbon dioxide, so one number should not be interpreted alone.

6.4 When heat changes an enzyme assay

MOLECULAR DIAGNOSTICS · A ROOM · CALLBACK · CHOICE

SCENE 30W

A separate reporter-enzyme assay runs normally after a 37-degree pre-incubation and loses most activity after the same enzyme is held at 55 degrees. Substrate, buffer and reading method are unchanged.

GUIDE 61W

Four options, and only one thing changed between the two runs. Ask of each option whether it names something that was actually varied. Substrate, buffer and readout are the same. The patient sample and the target sequence were not heated. So the answer has to be about the thing that was, and the two temperatures pull in opposite directions on it.

ASK 9W

What does the temperature comparison show about the catalyst?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a protein's function depends on it keeping its shape · a buffer is a chemical system that resists large pH changes when acid or base is added · cell structure and function — taken as read

VERDICT 80W

Enzyme activity depends on both molecular motion and protein structure. Modest warming can speed collisions, but enough heat can disrupt the shape that creates the active site. Here the same reporter enzyme, substrate, buffer and readout are used after two pre-incubation temperatures. The cleanest interpretation is loss of enzyme activity at the higher temperature. This is not a PCR cycling rule, and it does not

show that a target nucleic acid was destroyed. The experiment is specifically about the catalyst.

TAKEAWAY 15W

A controlled comparison can show which part of an assay is sensitive to changed conditions.

END OF THE STAGE 19W

Use homeostasis, gas exchange, circulation, oxygen transport, feedback, organ-system integration as an evidence chain rather than as isolated facts.

The Immune System Turns

PLAN CARD 105W

Day 18, and this is Ortiz's stage. Several patients improved for a week and are now deteriorating fast, including the paramedic who drove the first case in. The clinical immunologist, Yusuf Haddad, wants drugs that damp the immune system started tonight and Ortiz is not far behind him; Morales wants the timeline built first. Ortiz is right — people are dying on the ward while the argument runs. This stage you build that timeline anyway, because it is what tells you which drug. Haddad has marker levels for all six patients and dates against only half of them. The paramedic's family is in the corridor.

BRIEFING 20W

Several patients improve and then suddenly deteriorate as inflammation escalates. The player must distinguish protective immune responses from damaging dysregulation.

WHAT YOU DECIDE 20W

Construct a timeline of immune responses and decide whether deterioration reflects persistent agent burden, delayed adaptive response, or excessive inflammation.

7.1 Build the immune response timeline

IMMUNOLOGY & TREATMENT · A ROOM · SEQUENCE

SCENE 32W

Patients are improving and then getting worse again, and the immunology lab wants the normal response on the board before it says which part has gone wrong. Four cards, no order yet.

GUIDE 57W

These four are ordered by speed and by how specific they are, and that ordering is what makes a clinical measurement readable. Ask of each how long it takes and what it recognises. Pattern recognition fires within hours. Finding the right clone and expanding it takes days. And one of these is what is left over afterwards.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

the immune system has a fast general response and a slower specific one · cell structure and function — taken as read · pathogens: viruses and bacteria — taken as read

VERDICT 68W

The phases are ordered by speed and by specificity, and that ordering is what makes a clinical measurement interpretable. Innate sensors fire within hours because they recognise patterns rather than a particular agent. Local signals then recruit cells to where the damage is. Antigen-specific lymphocytes take days because they have to find the right clone and expand it. Memory is what persists after — a smaller population, kept.

TAKEAWAY 10W

Innate and adaptive immunity are coordinated phases, not competing systems.

7.2 Protective response or runaway inflammation?

CLINICAL PHYSIOLOGY · RIVERTON GENERAL — CLINICAL WING · A ROOM ·
DIAGNOSIS

SCENE 37W

Several patients began to worsen after the amount of detectable pathogen had already started to fall. The panel from the last 48 hours is on the screen, and the ward wants to know what it is treating.

GUIDE 58W

Five readings, and the timing is the discriminator. Ask of each candidate whether it would have the pathogen curve rising or falling while the patient worsens. If the agent were driving this, the two would climb together. Neutralising antibody being present says the specific response has arrived. The negative cultures remove the rival explanation for a late deterioration.

ASK 10W

Which mechanism best explains the reversal in the full panel?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

the immune response can damage tissue while it clears an infection · cell structure and function — taken as read · pathogens: viruses and bacteria — taken as read

VERDICT 68W

Timing is the discriminator. If the agent were driving this, its curve and the damage would rise together — instead the pathogen signal is falling while inflammatory signals climb and the patient gets worse. That is a response that has outlived its cause. Neutralising antibody being present says the specific response has arrived, and the negative bacterial cultures remove the rival that would otherwise explain a late deterioration.

TAKEAWAY 24W

A biological response can become part of the problem. The strongest explanation must account for timing as well as the direction of each measurement.

7.3 Treat the mechanism, not the marker

CLINICAL PHYSIOLOGY · RIVERTON GENERAL — CLINICAL WING · A PERSON · VALUE

SCENE 30W

Dr. Nia Okafor, the immunology lead, has one rapid laboratory slot before tonight's treatment conference. Patients are worsening late, and four measurements from archived samples could return on different timescales.

GUIDE 51W

One laboratory slot before tonight's conference, and four measurements that return on different timescales. Open each and ask what mechanism its result would establish, not how interesting the number is. Patients are worsening late, which narrows what a useful answer looks like. Buy the one that changes what the team treats.

ASK 14W

Which measurement most changes the mechanism the team should carry into tonight's treatment discussion?

BEHIND THE BUTTON 139W

Why late worsening is the clue. Deterioration after the peak of viral replication points away from the virus itself and towards the response to it. Those imply different treatments — antiviral versus immune-modulating — and giving the wrong one late is worse than giving nothing.

Why timescale is part of the value. A measurement that returns after the conference cannot change tonight's decision. That is not a reason to prefer fast and useless, but it does mean the slot has to be spent on something that both discriminates and arrives.

What a marker is and is not. A number that rises with severity tells you how ill somebody is. A number that identifies which process is driving the illness tells you what to do about it. Only the second is a mechanism, and only the second changes the treatment.

TAKES AS READ

the same bedside deterioration can arise from different mechanisms · a trend over time can distinguish mechanisms that overlap at one time point · innate and adaptive immunity — taken as read

VERDICT 88W

A single inflammatory value can be high while the agent is still driving disease or after the immune response becomes the main source of injury. The mechanisms separate when their trajectories separate. If agent burden falls while inflammatory markers continue rising, the host-driven explanation gains support. If both rise together, ongoing agent activity remains plausible. Cell-state profiling and pathology add useful detail, but they do not answer the urgent causal question as directly. The

most valuable measurement is the one that splits the live hypotheses before the decision.

TAKEAWAY 19W

The most valuable measurement is the one whose possible outcomes separate competing mechanisms that would lead to different actions.

7.4 Spend the first response reserve

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM · CALLBACK
· VALUE

SCENE 31W

Dr. Sofia Morales, the epidemiology lead, has one response slot she can start this afternoon. Four proposals are ready, and tomorrow's hospital reports will arrive before any second project can begin.

GUIDE 52W

One response slot you can start this afternoon, and four proposals. Open each and ask what tomorrow's conclusions would look like with and without it. Hospital reports arrive before any second project can begin, so this purchase decides what the team can reliably say tomorrow rather than what it will eventually know.

ASK 12W

Which first purchase changes what the response team can reliably conclude tomorrow?

BEHIND THE BUTTON 122W

Why the first purchase is different. It runs while nothing else does, so it sets the baseline everything later is compared against. A measurement started tomorrow can still be useful; one that should have started today can leave a gap in the record that nothing later fills.

What 'reliably conclude' rules out. Data that arrives without a comparison — cases with no denominator, positives with no sampling frame — cannot support a conclusion however large it is. The proposals differ mostly in whether they produce a comparison or only a count.

Why the hospital reports change the calculus. They arrive whatever you buy, so a proposal that duplicates what they will contain has spent the slot on something you were getting anyway.

TAKES AS READ

three hospitals can only be combined if they count cases the same way · an independent signal can be valuable without defining an individual case

VERDICT 83W

Three hospitals are about to produce three curves. Those curves can be combined only if the same operational definition created them. A standard line list preserves dates, symptoms and denominators in one comparable structure. Wastewater adds an independent population signal, but it cannot define a patient case. Expert impressions have no common denominator, and a warning adds no evidence. The first scarce action should make tomorrow's observations comparable; the independent streams become more useful once there is a clinical baseline to compare with.

TAKEAWAY 14W

The best first measurement is the one that makes later evidence comparable and decision-ready.

END OF THE STAGE 20W

Use innate immunity, adaptive immunity, antibodies, T cells, inflammation, immune regulation as an evidence chain rather than as isolated facts.

The Hidden Transmission Route

PLAN CARD 98W

Day 24, and Morales gets her stage. Ward isolation has held for nine days and cases keep appearing in neighbourhoods with nobody in hospital. A field epidemiologist, Grace Oduya, has pulled a sewer sample from the north of the city that was kept by chance and tested this morning — positive four days before the first patient walked in. This stage you decide where transmission is actually happening and where to put surveillance. Oduya has three sampling teams and the city has four watersheds. The nine days of ward isolation have already cost the hospitals their elective work.

BRIEFING 17W

Cases continue despite hospital isolation. Wastewater and neighborhood data suggest transmission is occurring before severe symptoms appear.

WHAT YOU DECIDE 17W

Identify the hidden route and target surveillance where it changes decisions rather than where sampling is easiest.

8.1 Match the surveillance signal

IMMUNOLOGY & TREATMENT • A ROOM • PROTOCOL

SCENE 32W

Hospital isolation has held for nine days and cases keep appearing. Four data streams are on the table at the wastewater plant, and each one saw a different slice of the city.

GUIDE 61W

Four data streams, and each one saw a different slice of the city. Pair them by asking who is in that sample, and what the stream can and cannot establish. A sewer signal does not name a person and does not prove waterborne spread. And prevalence counts everyone currently ill, which is why it can stay high while new cases fall.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

who ends up in a dataset is decided by how the data was collected • populations, transmission chains and exponential growth — taken as read

VERDICT 88W

Each surveillance stream samples a different population. Wastewater can detect community shedding without requiring people to seek care. It can therefore move before hospital admissions. It does not identify an individual, and a positive sewer signal does not prove waterborne transmission. Household interviews can reveal exposure timing but depend on recall and participation. A volunteer transit survey can be biased by who agrees to answer. Prevalence counts everyone currently ill, so it can remain high even after incidence — the number of new cases — begins to fall.

TAKEAWAY 22W

Surveillance streams observe different populations; timing differences can reveal hidden transmission, but wastewater detection does not by itself establish a transmission route.

8.2 How many by morning

CELL & MEMBRANE BIOLOGY · A ROOM · BALLPARK

SCENE 34W

The culture starts at 500 cells and the line divides every 40 minutes in this medium. The bench needs a number for eight hours from now, before it decides how much medium to prepare.

GUIDE 60W

Six numbers, and two of them are the same eight hours in different units, with the conversion between them offered as a third. Ask of each whether the exponent needs it. The shape matters more than the answer. Ninety more minutes adds two more doublings, which quadruples the total. The same arithmetic is frightening when the thing doubling is cases.

ASK 8W

Estimate the number of cells after eight hours.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

each division turns one cell into two

VERDICT 71W

8 hours is 480 minutes. That is 12 doubling times. 12 doublings is a factor of about 4,100, so 500 cells become about 2 million. The shape matters more than the number. Add 90 minutes and you add 2 more doublings, which quadruples the answer. That is why a culture left overnight runs out of medium. It is also why the same arithmetic is frightening when the thing doubling is cases.

TAKEAWAY 15W

Division doubles a population, so growth is exponential and its rate is a doubling time.

8.3 The same genome, different cells

CELL & MEMBRANE BIOLOGY · A PERSON · CHOICE

SCENE 36W

Staining shows the receptor on airway cells at high density and absent from the muscle cells two millimetres away — same patient, same section, same genome in every nucleus. Both tissues were fixed and stained together.

GUIDE 63W

Four options, and both tissues came from one patient and one section, stained together. Ask of each whether having a gene and using it are the same thing. Every nucleus holds the whole genome. So what has to differ is which genes are read, and that difference is what makes an airway cell an airway cell. It also decides which illness this becomes.

ASK 15W

Every cell has the same genome. Why do only some of them make the receptor?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

every cell in a body carries the same genome

VERDICT 82W

Every nucleus holds the whole genome, so having a gene explains nothing about where its protein appears. Gene regulation decides it: transcription factors and the state of the chromatin decide which genes are read in which cell type, and that regulation is what makes an airway cell an airway cell. For an outbreak it is the practical point — tissue tropism follows the expression pattern of the receptor, which is why this agent causes a respiratory illness rather than a muscular one.

TAKEAWAY 13W

Every cell carries the same instructions and reads a different part of them.

8.4 Clean side, dirty side

ECOLOGY & ONE HEALTH · ONE HEALTH FIELD STATION · A ROOM · BELT

SCENE 40W

The courier leaves in ten minutes and the tray has to be sorted first. One side of the bench has been touched by nothing outside the cabinet. The other has been open to the room, a glove, or a patient.

GUIDE 48W

Two bins, and the question is whether the item has been open to anything outside the cabinet since it was last sterile. If it has, it cannot go back to the clean side however clean it looks. Sort on where it has been, not on what it is.

ASK 11W

Send each item to the side its history puts it on.

BEHIND THE BUTTON 99W

Why the two sides never mix. Contamination is not visible, and it is not undone by looking careful. Once an item has been open to the room it can carry what the room carries, and putting it back on the clean side moves that into every later sample.

Why "it looks fine" is the wrong test. A sealed swab dropped on the floor is sealed and is on the dirty side. An open plate that has sat in a running cabinet for an hour is open and is on the clean side. The history decides it, not the appearance.

TAKES AS READ

an item can be sterile, and can stop being sterile without looking different

VERDICT 148W

The clean side is defined by history rather than by appearance, which is the whole discipline of aseptic work. An item that has been inside a running cabinet, or has never been unsealed, has been open only to filtered air and stays on the clean side. Anything that has met the room — a glove that touched a door, a tube that stood on the bench, a plate carried down the corridor uncovered — has been open to whatever the room carries, and looks exactly the same afterwards. That is why the rule is about where a thing has been and why a dropped sealed swab is dirty while an hour-old open plate in a cabinet is clean. Getting it wrong does not spoil one sample; it puts the room's organisms into every result the tray produces, which is a contamination nobody can distinguish afterwards from a finding.

TAKEAWAY 14W

What decides contamination is where a thing has been, not what it looks like.

8.5 What kind of agent fits the whole panel?

CLINICAL PHYSIOLOGY · RIVERTON GENERAL — CLINICAL WING · A ROOM ·
CALLBACK · DIAGNOSIS

SCENE 39W

Pathology has four results from the same positive specimens and still cannot name the agent class. The samples are waiting under the hood. Dr. Lena Ortiz needs a defensible broad classification before the diagnostic team commits its overnight run.

GUIDE 56W

Five readings, and three of them are negatives doing the classification. No cellular forms. No bacterial marker. No fungal wall. Ask of each candidate how many of the five it covers. The signal rises only with living host cells, and the extraction blank is clean. Note what this cannot reach: which nucleic acid, or which agent.

ASK 10W

Which explanation fits every reading, including the quiet negative controls?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

cells and viruses differ in whether they can copy themselves alone · a negative control is a result, not an absence of one

VERDICT 78W

The negative cellular tests do most of the classification work. No cellular forms are seen, the bacterial marker is absent, and the fungal-wall assay is negative. The signal rises only with living host cells, which supports an agent that depends on host machinery. A clean extraction blank makes laboratory carryover less likely. Together these observations support a virus-like class. They do not establish whether the genome is RNA or DNA, and they do not identify the exact agent.

TAKEAWAY 19W

Agent classification should combine cellular structure, host dependence, molecular evidence, and controls rather than rely on one striking image.

END OF THE STAGE 19W

Use incidence, prevalence, transmission chains, sampling, R concepts, asymptomatic spread as an evidence chain rather than as isolated facts.

The Animal Connection

BEFORE THE STAGE — CANVASS

Who in Ferry Ward is actually staying home

The transmission estimate assumes the advice was followed, and the difference between a ward that isolated and a ward that could not decides whether the curve is bending. Ask along the streets until enough answers have come back to tell those two apart.

PLAN CARD 105W

Day 33. Sequences from three patients sit close to a virus found in animals along the river, and the council has signed an order to clear the heron colony on Saturday. The wildlife vet, Mateo Castellanos, has been out there twice and says the herons are the wrong animal; the One Health lead, Arjun Singh, will not say either way until the sampling is designed properly. This stage you decide what role any animal is actually playing and design the campaign that could show it. What is being argued about is who infects whom over time, not which animal is nearby or easiest to blame.

BRIEFING 28W

Genetic similarities and field reports point toward an animal reservoir near the river corridor. The player must investigate without assuming that the most visible species is the source.

WHAT YOU DECIDE 12W

Build a defensible reservoir hypothesis using ecological exposure, prevalence, and genetic relatedness.

9.1 Reservoir, vector, or incidental host?

ECOLOGY & ONE HEALTH · ONE HEALTH FIELD STATION · A ROOM · PROTOCOL

SCENE 34W

Sequences from three patients sit close to a virus found in animals along the river corridor. The field station has four ecological patterns on the board and a trapper waiting outside with more animals.

GUIDE 56W

Four ecological patterns and four roles. Pair them by asking about dynamics rather than proximity. Does the agent persist in this species? Does the species pass it on? Does it stop there? One species is simply common near cases and always tests negative. Sharing a habitat with people is not the same as sharing an infection.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a species can carry an agent, spread it, or merely be near it · pathogens: viruses and bacteria — taken as read

VERDICT 85W

Ecological role is a claim about dynamics, not about proximity or about which animal people are frightened of. A reservoir carries the agent persistently and stays well enough to keep carrying it. A vector moves it between hosts without being the place it persists. A species that sickens and dies without passing it on is where transmission stops. And a species that is simply common near cases and never tests positive shares a habitat with people, which is not the same as sharing an infection.

TAKEAWAY 11W

Ecological roles are inferred from dynamics, not appearance or proximity alone.

9.2 Test the reservoir hypothesis

ECOLOGY & ONE HEALTH · ONE HEALTH FIELD STATION · A PERSON · SEQUENCE

SCENE 37W

Trappers report sick animals on the flood plain, and nobody has yet sampled in a way that could tell a reservoir from a bystander that caught it from people. Four steps, and the campaign gets funded once.

GUIDE 62W

All four steps will happen and the campaign is funded once. A positive animal is not a reservoir. What turns it into one is a design that could have found the agent elsewhere and did not. Ask of each step what makes the next one interpretable. Exposure patterns say where to sample. Comparison sites are what make a prevalence figure mean anything.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

where you look decides what you can find

VERDICT 81W

A positive animal is not a reservoir. What turns it into one is a design that could have found the agent somewhere else and did not, which is why the order matters: exposure patterns say where to sample, and sampling built around comparison sites is what makes a prevalence figure mean anything. Only then do prevalence and genetic relatedness across hosts say whether this species is in the transmission cycle, and only then can repeated spillover be tested rather than assumed.

TAKEAWAY 9W

A reservoir claim requires population and evolutionary evidence together.

9.3 Choose the One Health campaign

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM · ALLOCATE

SCENE 30W

Dr. Arjun Singh, the 1 Health lead, has 1 field campaign and 20 sampling slots. Candidate species, comparison sites and a second time point all compete for the same crews.

GUIDE 53W

One campaign, twenty sampling slots, and more candidate species, sites and time points than the crews can cover. Watch the questions list rather than the map: it shows which comparisons your current plan can still make. A plan that samples the most animals and preserves no comparison cannot test the hypothesis at all.

ASK 12W

Which sampling plan preserves the comparisons needed to test the reservoir hypothesis?

BEHIND THE BUTTON 142W

Why comparisons rather than counts. Finding the pathogen in a species proves it can be infected, not that it is the reservoir. That claim needs a contrast: this species against others, this site against a comparison site, now against later. Slots spent without a contrast produce a positive result that settles nothing.

What a comparison site is for. Detection everywhere means the finding is not specific to the outbreak area. Detection here and not there is what links the reservoir to the human cases, and it costs half the campaign to be able to say it.

Why the second time point is expensive and hard to give up. Reservoir dynamics change with season and with the outbreak itself. One visit gives a snapshot that cannot distinguish an established reservoir from a passing spillover, which is exactly the question the response needs answered.

TAKES AS READ

a sample represents the population it was drawn from · a reservoir claim needs evidence that can distinguish persistence from spillback · pathogens: viruses and bacteria — taken as read

VERDICT 89W

A positive animal near a human case is compatible with several stories: reservoir, spillback from people, or shared exposure. The design has to create comparisons that separate those stories. Sampling the same species away from cases tests geography. Sampling other species in the same habitat tests species specificity. Returning later tests persistence. Sequencing can then compare what the field design actually found, but it cannot rescue a biased sample. Spending the campaign on one feared species near one household produces a precise result that cannot establish the population process.

TAKEAWAY 20W

A field design is defined by the alternative explanations it can rule out with the samples it chooses to collect.

9.4 Choose the membrane mechanism

CELL & MEMBRANE BIOLOGY • A ROOM • CALLBACK • PROTOCOL

SCENE 35W

Four observations from the culture bench each describe something moving across a membrane. The cell biologist wants each one named by its mechanism before the entry model goes further, without guessing from the molecule names.

GUIDE 58W

Four observations and four mechanisms, and the name of the molecule tells you nothing. Pair them on three tests. Which way does it go, downhill or uphill? Is a protein involved, or does it cross the lipid directly? And is it too big for any channel? The ATP being consumed is the receipt for one of those answers.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a concentration gradient has a direction • ATP is the cell's energy currency • cell structure and function — taken as read

VERDICT 73W

Three constraints tell these apart and none of them is the name of the molecule. Which way it goes: downhill needs no energy, uphill does, and the ATP consumption is the receipt. Whether a protein is involved: a small nonpolar molecule crosses the lipid directly, an ion cannot and needs a pore. And how big it is: a bound particle too large for any channel is taken in by the membrane wrapping it.

TAKEAWAY 12W

Transport mechanisms are inferred from constraints, not memorized from molecule names alone.

END OF THE STAGE 20W

Use reservoirs, vectors, food webs, host range, ecological sampling, One Health as an evidence chain rather than as isolated facts.

The Mutation

PLAN CARD 100W

Day 45. A variant with 3 changes has gone from 8 to 31% of sequences in 4 weeks, in 1 district, and the briefing is in an hour. Bergström will not call it selection yet. This stage you decide whether that rise means the variant is better at spreading, and get the frequencies onto a footing you can defend. The district that produced the rise also changed who it swabs, which Bergström noticed on Tuesday. The minister's briefing note already calls it a new strain. If you call selection on an artefact, the next real signal will be dismissed too.

BRIEFING 28W

A new genetic variant rises rapidly in one part of the city. The player must decide whether its increase reflects natural selection, random founder effects, or biased sampling.

WHAT YOU DECIDE 16W

Distinguish change in frequency from proof of advantage and identify evidence that tests competing evolutionary explanations.

10.1 From variation to evolutionary change

MOLECULAR DIAGNOSTICS · A ROOM · SEQUENCE

SCENE 35W

A genetic variant is rising in one part of the city and the word evolution is being used in the briefing room. Four cards describe how a population changes. Nobody has put them in order.

GUIDE 56W

These four are one chain, and nothing in it says a rising variant must be advantageous. Ask of each card what has to exist before it. Selection does not create the variant it needs when it needs one. And take the middle step away: frequencies still move, for reasons that have nothing to do with fitness.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

offspring inherit traits from their parents · mutation and genetic variation — taken as read

VERDICT 79W

Nothing in this chain says a rising variant must be advantageous, and that is the point of laying it out. Variation has to exist before anything can act on it — selection does not create the variant it needs when it needs one. Conditions then make some variants reproduce more than others, those variants leave more descendants, and the frequency moves. Take away the middle step and frequencies still move, for reasons that have nothing to do with fitness.

TAKEAWAY 10W

Natural selection changes frequencies through heritable differences in reproductive success.

10.2 Selection or sampling artifact?

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM ·
DIAGNOSIS

SCENE 35W

A variant has gone from 8 to 31% of sequences in 4 weeks and the briefing is in an hour. The evolutionary dynamics group has run its checks and put them all on one screen.

GUIDE 50W

Five readings, and each rules out a different rival. The rise repeats across hospital systems. Sequencing depth is stable. Neutral variants stay flat. Ask of each candidate how many of the five it survives. And note what even the best answer here does not measure: the size of any advantage.

ASK 11W

Which explanation best fits the frequency trend and the independent checks?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

a change in who gets tested changes what the tests find · mutation and genetic variation — taken as read

VERDICT 85W

A rising frequency is an observation, not a direct measurement of fitness. Selection is one explanation, alongside founder effects, changing sampling and technical bias. The checks here weaken those alternatives. The rise repeats across hospital systems, sequencing quality is stable, neutral control variants stay comparatively flat, and the trend survives adjustment for place and date. Together that makes selection the leading explanation in this panel. It does not measure the size of a transmission

advantage. Repeated growth-rate comparisons or functional evidence would strengthen that claim.

TAKEAWAY 16W

A changing share needs competing explanations tested before it can support a claim about biological advantage.

10.3 Track a rising variant

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON · BALLPARK

SCENE 39W

The variant briefing starts in 30 minutes. Week one and week three used different sample totals, and the surveillance hub needs the actual change before anyone argues about why it happened. The raw counts are still on the board.

GUIDE 61W

Six numbers, and two of them are the weeks pooled together, which answers a different question. Ask of each whether it belongs to week one or week three. A frequency is a ratio, and it only means something if both denominators were produced the same way. A change in who gets sequenced moves the numerator with nothing happening in the city.

ASK 9W

Estimate the change in variant frequency, in percentage points.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a fraction needs both a numerator and the denominator it came from · dNA structure and replication — taken as read

VERDICT 35W

A frequency is a ratio, and the ratio only means something if the two denominators were produced the same way. A change in who gets sequenced moves the numerator with nothing happening in the population.

TAKEAWAY 11W

Evolutionary inference begins with accurate frequencies and then tests causal explanations.

10.4 Separate binding from entry

CELL & MEMBRANE BIOLOGY · A ROOM · CALLBACK · CONTROL

SCENE 32W

Dr. Maya Chen, the cell biology lead, has a simulated labelled-particle entry assay on vulnerable airway cells. Three perturbations are ready, and the therapy meeting needs evidence about the first physical step.

GUIDE 66W

The number you are watching is the entry signal: how much labelled particle ends up inside the cells. Three things can be changed, and only one of them acts on the step the meeting is asking about. A response counts only if it is larger than the well-to-well scatter of the assay. Change one, run it, restore it, and name the change the entry signal follows.

ASK 17W

Which single change makes the entry signal follow the receptor interaction rather than exposure or antibody handling?

BEHIND THE BUTTON 166W

What the assay measures, and what it does not. The label reports particles that have got inside the cell, not particles stuck to the outside. That distinction is the whole point of the experiment: binding to the surface and entering the cell are separate steps, and a therapy that blocks one may do nothing about the other.

Why a matched antibody is in the list. It is there to be changed and produce nothing. An antibody of the same class that does not target the receptor controls for everything handling an antibody involves — the buffer, the incubation, the washing — so a response to the real blocker means something only if this one gives none.

Why the dose is in the list too. Give more particles and more get in, which moves the number without saying anything about the mechanism. It is the plausible confound: a change that moves the readout for a reason that has nothing to do with the step being asked about.

TAKES AS READ

a causal test changes one factor while holding the others fixed · a reversible effect is stronger evidence than a one-way coincidence · transcription, translation and gene expression — taken as read

VERDICT 83W

Receptor abundance is only an indirect clue. A causal test asks whether the proposed interaction is required for entry. Blocking that interaction while keeping dose and handling fixed sharply lowers the entry signal. A matched antibody leaves the signal unchanged, which controls for adding an antibody at all. Washing the blocker away restores entry, so the effect follows the manipulated link in both directions.

Doubling exposure changes signal size but does not explain why the receptor-specific perturbation removes entry under otherwise matched conditions.

TAKEAWAY 19W

A causal mechanism is stronger when changing one proposed link changes the response and reversing that change restores it.

END OF THE STAGE 19W

Use mutation, selection, genetic drift, founder effects, fitness, allele frequency as an evidence chain rather than as isolated facts.

Treatment Failure

PLAN CARD 103W

Day 60. The therapy that bought this city six weeks is failing in one patient in four, and the failures cluster on the wards where it has been used most. Ortiz wants the dose raised today. Bergström has a resistant sequence from a river sample banked a month before the first dose was given anywhere. This stage you work out why it is failing and decide what to do that does not make it worse. Raising the dose is the fastest thing this room can do and might also be the worst. Nobody has yet checked whether the failing patients received every round.

BRIEFING 30W

A therapy that initially helped is failing in a growing subset of patients. The player must determine whether the cause is biological resistance, poor delivery, incorrect dosing, or patient heterogeneity.

WHAT YOU DECIDE 18W

Separate inherited resistance from non-genetic treatment failure and propose a strategy that reduces selective advantage without undertreating patients.

11.1 Why did the therapy fail?

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM ·
DIAGNOSIS

SCENE 38W

A therapy that worked is failing, and another patient has deteriorated overnight. The ward has drug levels, burden over time, the target-site sequence and a reference isolate. Pharmacy wants a new plan before the afternoon doses are prepared.

GUIDE 58W

Five readings, and the normal ones do the work. The drug reached a working concentration. The burden fell. The administration record is complete. Ask of each candidate how many of the five it covers. What happened next is a rebound while the patient was still taking it, and the reference isolate staying sensitive says the assay is fine.

ASK 7W

Which explanation fits the complete treatment-response panel?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

a drug only works if it reaches the target at a high enough concentration

VERDICT 89W

Treatment failure has several causes and they are told apart by which measurements are normal. The drug reached a concentration that works and the burden fell, so it was working — that rules out delivery and dose. It then rebounded while the patient was still taking it, and the resistance-associated variant at the target site became dominant over the same period. The reference isolate staying sensitive is what says the assay and the drug are both still fine. Evolution is visible here on the timescale of a ward round.

TAKEAWAY 16W

Evolution can be observed in real time when a treatment changes which variants are most successful.

11.2 How resistance spreads

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON · SEQUENCE

SCENE 39W

The therapy worked for six weeks and now fails in one patient in four. Failures cluster on the wards where it has been in use longest, and somebody in the meeting has said the organism learned to resist it.

GUIDE 56W

These four are one chain, and somebody in the meeting has said the organism learned. Ask of each card what has to be true before it. The variation is already there, rare, before the drug arrives. What the drug changes is which variants leave descendants. The clustering on long-treated wards is that process having run longer.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a population contains variation before anything acts on it · pathogens: viruses and bacteria — taken as read

VERDICT 74W

Treatment does not teach an organism anything, and it does not create the variant it needs. The variation is already there, rare, before the drug arrives. What the drug changes is which variants leave descendants: it removes the susceptible ones more effectively, the less susceptible survive and reproduce, and the treated population ends up enriched for resistance. The clustering on long-treated wards is that process having run longer, not the organism having learned faster.

TAKEAWAY 13W

Selection acts on existing or newly arising variation by differential survival and reproduction.

11.3 Recover treatment effectiveness

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON · CHOICE

SCENE 35W

Effectiveness has fallen from 82 to 51% in 9 weeks. 11 failures had dosing on record and trough levels in range. The ward wants one decisive test before tomorrow's treatment protocol is rewritten for everyone.

GUIDE 57W

Four options, and a treatment can fail in three places: given wrongly, absorbed poorly, or no longer working at the target. Ask of each which of the three it tests. Documented dosing and in-range troughs weaken the first two. And note what makes the fourth option a test rather than an observation: matched successes are the control.

ASK 7W

One thing before anybody's prescription changes. Which?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a matched comparison is what turns a finding into evidence · pathogens: viruses and bacteria — taken as read

VERDICT 86W

Work through where a treatment can fail. It can be given wrongly, absorbed poorly, or reach the target and no longer work there. Documented dosing and trough concentrations in the expected range weaken the first two explanations. That makes a change at the target more plausible, especially with effectiveness falling from 82 to 51% over 9 weeks. Sequencing failures beside matched successful cases tests whether a resistance-associated variant tracks the failures rather than merely

existing. Matched successes are the control that turns association into stronger evidence.

TAKEAWAY 12W

A resistance response should combine evolutionary evidence with pharmacology and patient safety.

11.4 From genetic information to protein

MOLECULAR DIAGNOSTICS · A ROOM · CALLBACK · SEQUENCE

SCENE 45W

Affected cells are making RNA and proteins the sequencing centre does not recognise. Four cards describe how a cell gets from a gene to something that does a job. Nobody can say where the change is until the chain is on the board in order.

GUIDE 60W

These four are one chain, and abundance can change at any stage independently. That is why the order is worth stating rather than treating expression as one event. Ask what each card needs to exist first. A template has to be reachable. A transcript has to exist. And a chain of amino acids does nothing at all until it folds.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

genes are read into RNA and RNA is read into protein · dNA structure and replication — taken as read

VERDICT 72W

Abundance can change at any one of these stages independently, which is exactly why the chain is worth stating rather than treating expression as one event. The template has to be reachable before anything transcribes it; a transcript has to exist before a ribosome can read it; and a chain of amino acids does nothing at all until it folds. Each stage is a separate place for the answer to be hiding.

TAKEAWAY 13W

Changes in protein abundance can arise at several distinct stages of gene expression.

Use resistance, selection pressure, dose-response, targets, pharmacokinetics, heterogeneity as an evidence chain rather than as isolated facts.

Design the Intervention

BEFORE THE STAGE — EVADE

The journalist at the gate who wants a number

A reporter has been at the main gate since six wanting today's case count before it is confirmed, and Morales, the epidemiology lead, has been clear that nothing goes out before the reconcile. Keep clear and finish your round. A number given at a gate is a number the hospital will spend a week correcting.

PLAN CARD 109W

Day 78. The city wants an intervention and the room is arguing as though it can only have one. The antibody everyone favours binds the exact stretch of protein that mutated in the north, which Bergström has now said three times. This stage you choose what to fund and say what evidence is needed before any of it reaches a person. The options also arrive on different timescales — one is available on Monday, one in a year — which is the whole argument for layering them rather than picking a favourite. Back a single glamorous option and the city has one point of failure against something still changing.

BRIEFING 30W

The city needs an intervention that can be manufactured, tested, and deployed. The player must compare vaccines, targeted therapies, and nonpharmaceutical measures using evidence about mechanism and evidence about populations.

WHAT YOU DECIDE 12W

Select a layered intervention portfolio and identify the evidence required before deployment.

12.1 Choose the intervention layer

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM ·
PROTOCOL

SCENE 36W

The city wants an intervention and the room is arguing as though only one can be chosen. Four proposals are on the table and each acts at a different moment in the course of an infection.

GUIDE 49W

Four proposals and four moments in the course of an infection. Pair them by asking when each one acts: before exposure, at entry, after the agent is established, or between people. Different mechanisms fail in different ways, which is the whole argument for layers rather than for a winner.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

an infection has a before, a during and an after, and different tools act at each

VERDICT 85W

Each class acts at a different point, which is why a layered strategy is stronger than a bigger version of any one of them. Blocking entry acts before or during establishment of infection. An anti-inflammatory acts later on host injury rather than directly on the agent. Vaccination builds immune memory before exposure and can reduce infection or severe disease, depending on the vaccine and agent. Ventilation, isolation and contact reduction work at the population level. Different mechanisms create different strengths, timing constraints and failure modes.

TAKEAWAY 15W

A layered strategy is stronger when each intervention has a defined mechanism and failure mode.

12.2 From candidate to justified trial

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON · SEQUENCE

SCENE 39W

Two compounds show activity in cell culture and the city wants one of them in patients within the month. Neither has been through an animal model, and the protocol on the table names no endpoint and no stopping rule.

GUIDE 63W

All four steps will happen and urgency changes their speed, not their order. Ask of each what the next step would mean without it. A target says what the compound is supposed to do. Animal evidence says whether it does anything, and what it does to the animal. Without an endpoint and a stopping rule, a trial can neither conclude nor be halted.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

activity in a dish is not evidence of benefit in a person

VERDICT 81W

Urgency changes the speed of this path and not its structure. A target and a mechanism come first because they say what the compound is supposed to do; animal and laboratory evidence says whether it does anything and what it does to the animal; the trial design says what would count as working and when to stop. Skip the endpoint and the stopping rule and you have a trial that cannot conclude anything and cannot be halted when it should be.

TAKEAWAY 14W

Urgency changes speed and coordination, not the need for interpretable evidence and participant protection.

12.3 Build a layered intervention portfolio

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON ·
ALLOCATE

SCENE 30W

Dr. Nia Okafor, the immunology lead, has six plausible intervention layers and only three funding slots. Some can work this month; others take longer or share the same failure mode.

GUIDE 49W

Six plausible layers, three funding slots. Watch which response questions the current portfolio can still answer. Layers that share a failure mode look like two defences and behave like one, so a portfolio that holds if one assumption fails is worth more than three strong layers that fail together.

ASK 12W

Which three-layer portfolio still answers the response needs if one assumption fails?

BEHIND THE BUTTON 121W

What a layered intervention means. No single measure works perfectly, so the strategy is several partial ones whose failures are independent. Two layers that both depend on people reporting symptoms promptly are not two layers; they are one, bought twice.

Why timing separates them too. Some layers work this month and some take a season to stand up. A portfolio entirely made of slow layers leaves the next four weeks uncovered, whatever it eventually achieves.

How to test a portfolio. Take each assumption in turn — people comply, supply arrives, the pathogen behaves as modelled — and ask what is left standing if it is wrong. A portfolio that survives every single failure is the one that keeps the response working.

TAKES AS READ

different mechanisms can fail for different reasons · a portfolio gives up some good options because its resources are finite

VERDICT 86W

Layered control is stronger when the layers do not all wait for the same event or fail for the same reason. Ventilation reduces exposure opportunities without depending on who has been diagnosed. Rapid isolation acts after detection and shortens opportunities for onward spread. A vaccine takes longer, but it adds a distinct immune mechanism for the future. Treatment and paid leave can also matter, but the fixed three-slot portfolio cannot buy everything. The point is to preserve different mechanisms and timescales under a real resource constraint.

TAKEAWAY 18W

A strong intervention portfolio combines different mechanisms and timescales instead of simply taking the three largest headline effects.

12.4 Negative, while the door keeps opening

IMMUNOLOGY & TREATMENT · A ROOM · HOLD

SCENE 43W

The isolation room has to stay at negative pressure to the corridor, so that air moves in rather than out. Staff come and go, the anteroom door is propped during a turn, and the extract fan is the only thing that answers it.

GUIDE 60W

Hold the room's pressure difference inside the band on the gauge. The band narrows as the shift goes on, because the ward fills and a room that is marginal with two patients is not acceptable with six. The extract fan is your control, and every door that opens is a leak that goes on leaking until the fan answers it.

ASK 11W

Hold the isolation room below the corridor while the shift runs.

BEHIND THE BUTTON 152W

What negative pressure is for. Air moves from higher pressure to lower, so a room held below the corridor draws air in under its door rather than pushing it out.

Everything airborne in that room stays in it and leaves through the filtered extract.

Why an open door is not a step in the reading. Propping the anteroom door does not lower the difference by two pascals and stop. It opens a path that keeps bleeding pressure for as long as it stands open, so a room left alone does not settle — it drifts toward the corridor and then past it.

Why a positive room is worse than no room. Once the difference goes the other way the room is pushing its air into the corridor, and it goes on doing it silently. The gauge is the only thing that says so, which is why it is read rather than assumed.

TAKES AS READ

air moves from higher pressure toward lower pressure

VERDICT 157W

Every event here is a step in the rate rather than a step in the reading. A propped door does not drop the difference by a fixed amount; it opens a path that bleeds pressure for as long as it is open, so a room left alone drifts and keeps drifting. That is why the extract fan has to be moved to a new setting and left there rather than nudged and released. The band narrows because the ward fills: a difference that is marginal with two patients in the room is not acceptable with six, and the same fan has to be held closer as the afternoon goes on. And the failure mode is quiet — a room that has gone positive is pushing its air into the corridor and looks exactly like a room that has not, which is why the gauge is on the wall and why somebody is made responsible for reading it.

TAKEAWAY 13W

A leak you do not answer is not a leak that closes itself.

12.5 Locate the molecular change

MOLECULAR DIAGNOSTICS · A ROOM · CALLBACK · PROTOCOL

SCENE 33W

Four patterns from the gene-expression bench each pair one measurement with another. The cell biologist wants each pattern placed at the stage it reports on before anyone proposes a host pathway to target.

GUIDE 63W

Four patterns on the left, and four places the change could sit. Pair them by asking which two measurements each pattern compares, and which direction they disagree in. Message up with protein flat is one end. Protein up with message flat is the other. And a protein present in normal amounts but not working is a claim about the molecule, not the quantity.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

the amount of a protein and the amount of its message are measured separately ·
dNA structure and replication — taken as read

VERDICT 87W

RNA abundance, protein abundance, sequence and activity are four different measurements, and the central dogma is a chain of transformations rather than a promise that they move together. Message up with protein flat means the message is not being turned into protein, or the protein is being destroyed as fast as it is made. Protein up with message unchanged means the opposite end. And a protein present

in normal amounts but not working is a claim about the molecule, not about how much of it there is.

TAKEAWAY 17W

The central dogma is a chain of measurable transformations, not a claim that every stage changes together.

END OF THE STAGE 20W

Use antigens, immune memory, drug targets, intervention layers, efficacy versus effectiveness as an evidence chain rather than as isolated facts.

The Trial Signal

PLAN CARD 98W

Day 96, and the trial has reported: 10 hospitalisations in the treated group against 20 in the control, out of 400 people. Most of that benefit sits at the one site that enrolled the youngest patients, and the serious adverse events are not spread evenly either. The health minister wants a number before deciding how much to buy, and has been given one percentage she would like confirmed. This stage you read the result properly and decide whether the trial continues, expands or stops. Nobody in the room has yet said how many people 20 is out of.

BRIEFING 31W

Early trial results appear encouraging, but enrollment is small and adverse events are uneven. The player must distinguish a real effect from random variation and decide whether the trial should expand.

WHAT YOU DECIDE 20W

Make a conditional trial decision that weighs benefit, uncertainty, and safety rather than relying on one p-value or headline percentage.

13.1 Did the comparison stay fair?

CLINICAL PHYSIOLOGY · RIVERTON GENERAL — CLINICAL WING · A ROOM · CHOICE

SCENE 38W

One large trial site enrolled much younger patients than the others. It also supplies much of the apparent benefit. The original analysis plan already includes site-stratified estimates and a treatment-by-site check. The health ministry wants an answer today.

GUIDE 57W

Four ways to handle one imbalanced site. Ask of each whether the choice was made before or after the outcomes were seen. Randomisation makes groups comparable in expectation, not identical in every subgroup. Two of these options would let the analysis depend on the result. The plan already contains the tools, which is what makes this decidable.

ASK 16W

How should the team handle the site imbalance without choosing the analysis after seeing the outcome?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

randomization balances groups in expectation, not perfectly inside every site · an analysis chosen after seeing outcomes can introduce bias

VERDICT 78W

Randomization makes treatment groups comparable in expectation, not identical in every subgroup. A site that enrolled younger participants can create a pattern worth checking. That matters especially when the site carries much of the apparent effect. Use the site-stratified and interaction analyses specified before outcomes were seen.

Then inspect consistency and data quality, and report the heterogeneity. Dropping the inconvenient site or inventing a new subgroup after seeing the result would make the analysis depend on the outcome.

TAKEAWAY 17W

Randomization protects the treatment comparison on average; an imbalanced site pattern should be examined with pre-specified analyses.

13.2 Read effect size and uncertainty

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM · BALLPARK

SCENE 33W

The trial has reported, and the health minister wants a number before deciding how much to buy. The statistics office has the event counts for both arms and nothing has been converted yet.

GUIDE 56W

Six numbers, and two of them are the trial pooled together, which answers nothing here. Ask of each which arm it belongs to. And note that one result can be told three ways. A halved risk sounds enormous. Five fewer admissions per hundred sounds modest. Only the last two tell a minister how much to buy.

ASK 13W

Estimate the number of patients who must be treated for one to benefit.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a risk is events divided by the people who could have had them

VERDICT 66W

The same result can be told three ways. Risk halved sounds enormous. Five fewer hospital admissions in every hundred sounds modest. 20 patients treated for 1 to benefit sounds like work. All three are true, and only the last two tell a minister how much of the drug to buy. Reporting the relative figure alone is the commonest way a real but small effect is oversold.

TAKEAWAY 13W

Relative and absolute effects answer different decision questions and should be reported together.

13.3 Continue, expand, or pause?

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON · CHOICE

SCENE 35W

The estimated benefit is 24%, but the interval runs from 3 to 41. Serious adverse events are seven versus two. Enrolment is paused. The independent safety board meets tonight, with hospitals waiting for its decision.

GUIDE 61W

Four decisions, and two facts have to be held together: an interval from 3 to 41 per cent, and seven adverse events against two. Ask of each option whether it treats either number as more decisive than it is. Excluding zero is evidence against no effect, not a measurement of the effect. Raw counts do not separate chance from a signal.

ASK 13W

Promising, imprecise, and the adverse events are uneven. What does the board decide?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a wider interval means less is known, not that the effect is smaller

VERDICT 88W

A confidence interval from 3 to 41% is compatible with benefit across a wide range of sizes. Excluding zero under the planned model is evidence against no effect, not proof that the exact benefit is known. Seven serious adverse events versus two may be chance or a safety signal; raw counts alone do not decide which. Precision depends on sample size, event frequency and design. The board should apply the pre-specified safety boundary and event review. It can then resume, modify, expand or stop according to that plan.

TAKEAWAY 23W

Benefit and safety must both be judged through the pre-specified monitoring plan; a positive benefit estimate does not override a possible safety signal.

13.4 Find the controlling stage

MOLECULAR DIAGNOSTICS · A ROOM · CALLBACK · CHOICE

SCENE 39W

A familiar human protein is fortyfold more abundant in affected cells than in matched controls. Its messenger RNA has not been quantified, the core can fund two measurements, and the team is already arguing about which host pathway changed.

GUIDE 64W

Four measurements, and the protein result is compatible with all of the mechanisms. Ask of each option where it sits in the chain: upstream of the others, or downstream. Two of these test what happens to a transcript that nobody has counted yet. The core can fund two measurements, and the first should be the one that tells you where to spend the second.

ASK 11W

Abundance alone fits every mechanism. Which measurement do you make first?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a protein's abundance is set by how fast it is made and how fast it is destroyed · dNA structure and replication — taken as read

VERDICT 84W

A fortyfold protein increase is compatible with several mechanisms: more messenger RNA, more protein made from each RNA molecule, or slower protein degradation. The first measurement should sit upstream of the other two. Quantifying messenger RNA over the same time course tests whether transcript abundance moved with the protein. If RNA also rises strongly, increased transcription becomes a leading contributor. If RNA stays flat, the investigation moves downstream to translation

efficiency and protein turnover instead of pretending the protein result already identified the cause.

TAKEAWAY 12W

Protein abundance alone cannot distinguish increased transcription, changed translation, or altered degradation.

END OF THE STAGE 21W

Use randomization, control groups, effect size, confidence intervals, adverse events, stopping rules as an evidence chain rather than as isolated facts.

Containment or Adaptation?

PLAN CARD 109W

Day 121 districts are on visibly different paths and the council wants 1 policy by Friday. 90 of the region's hundred and 20 critical-care beds are full and severe cases are arriving at about 8 a day. Ortiz wants the beds protected; Morales wants the growth stopped where it is still cheap. Ortiz has the bed numbers by hour, Morales has the district growth rates, and the two documents have never been in the same room. This stage you build the arithmetic behind the decision and choose a policy that survives being wrong. The honest answer to an uncertain future is a set of triggers rather than one guess.

BRIEFING 23W

The outbreak is geographically uneven. Leaders must choose between aggressive local containment, broader mitigation, and long-term adaptation while the biological parameters remain uncertain.

WHAT YOU DECIDE 12W

Use scenarios and trigger-based policies instead of pretending one forecast is certain.

14.1 Build a decision model

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM · SEQUENCE

SCENE 33W

Three districts are on visibly different trajectories and the council wants one citywide policy by Friday. The modelling team has nine days of case data and a dozen parameters it can only bound.

GUIDE 56W

All four steps will be done, so ask what each one makes possible. A model is a structured argument, and its assumptions are as much of the output as its numbers. Built without a decision, it answers whichever question was easiest. And a scenario not tied to something observable is a scenario nobody will act on.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a model's output is only as good as the assumptions it was given

VERDICT 84W

A model is a structured argument, not an oracle, and its assumptions are as much of the output as its numbers. Stating the decision first is what stops it modelling something nobody asked about — a model built without a decision answers whatever question was easiest. The structure and the inputs come next, then the range of plausible worlds rather than one, and finally the triggers, because a scenario that is not tied to an observable is a scenario nobody will ever act on.

TAKEAWAY 9W

Models are structured arguments whose assumptions should remain visible.

14.2 Hospital capacity threshold

CLINICAL PHYSIOLOGY · RIVERTON GENERAL — CLINICAL WING · A ROOM · BALLPARK

SCENE 36W

The region has 120 staffed critical-care beds, and 90 are already full. Severe cases are arriving at about eight a day. The transfer centre is calling neighbouring hospitals while the council asks how many days remain.

GUIDE 52W

Five numbers, and two of them belong to other questions: the total staffed beds, and the average stay. Ask of each whether this bound depends on it. The answer assumes nobody leaves, which is the point of a bound. What it shows is what is missing: the discharge rate nobody is collecting.

ASK 12W

Estimate the upper bound on time to capacity, assuming nobody is discharged.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a bound is what you can say before the missing number arrives

VERDICT 26W

A bound is worth computing early because of what it shows is missing. Here it says the decision turns on a discharge rate nobody is collecting.

TAKEAWAY 9W

Simple bounds can reveal which missing data are decision-critical.

14.3 Choose an adaptive policy

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON · TRIGGER

SCENE 28W

Dr. Sofia Morales, the epidemiology lead, has current critical-care occupancy at 75% and a short worst-case trace from the capacity model. Two reversible actions each need lead time.

GUIDE 72W

One rule, written before the trace is released and fixed once it is. Overflow staffing and transfer planning take a day from the moment they are called, and at 95 % occupancy there is no staffed bed for the next admission. So the line goes back from 95 by however far occupancy climbs in a day — and not so far back that the unit is surged on an ordinary bad week.

ASK 16W

At what critical-care occupancy do you activate overflow staffing, given it takes a day to arrive?

BEHIND THE BUTTON 137W

Why precommit at all. Occupancy is at 75% and climbing, and every hour of deliberation is an hour of lead time spent. A rule set now is the same decision made calmly, and it is the only version that can be explained afterwards to people whose care was affected.

Why the two actions need different numbers. They take different times to have effect. A threshold set for the slower one triggers the faster one far too early; set for the faster one, the slower cannot finish in time. The lead time is the whole reason there are two.

Why reversible matters. Both actions can be undone if the trace turns out to be pessimistic, which is what makes it reasonable to commit to them in advance. An irreversible action would need a different standard of evidence entirely.

TAKES AS READ

a trigger is written before the next number arrives · reversible preparations still need time to take effect

VERDICT 86W

A trigger is a decision rule written before the next value arrives. That matters because preparation takes time even when the action is reversible. In the source numbers, 90 of 120 beds are already occupied, so the system starts at 75%. The no-discharge bound reaches critical occupancy quickly. Precommitting a staffing threshold and a later surge-capacity threshold prevents hindsight from changing the rule after every update. The policy remains adaptive because it responds to observations, but the standard for acting was fixed before those observations appeared.

TAKEAWAY 16W

Adaptive policy works best when observable thresholds and lead times are committed before the evidence moves.

14.4 Build a trustworthy amplification run

MOLECULAR DIAGNOSTICS · A ROOM · CALLBACK · SEQUENCE

SCENE 43W

Hospitals want a diagnostic by sunrise and the sample-processing room has four hours. Four steps are on the whiteboard. Amplification multiplies whatever is in the tube by a factor of 1 billion, which is the reason this order is argued about at all.

GUIDE 56W

All four steps will happen, so ask what each one protects. The amplification multiplies whatever is in the tube, including whatever came from the room. Identity has to survive extraction or the result belongs to nobody. And controls have to be in the run rather than added afterwards, which is what makes this an ordering question.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

amplification copies whatever template is present, wanted or not · DNA structure and replication — taken as read

VERDICT 79W

A patient result is the output of a controlled process, and read before the controls are read it is an instrument reading rather than evidence. Identity has to survive extraction or the result belongs to no one. The controls have to be in the run, not added afterwards — a positive control shows the chemistry worked, and a no-template control shows the signal did not come from the room. Only when they behave is the patient well worth interpreting.

TAKEAWAY 15W

A diagnostic result is the output of a controlled process, not simply a machine reading.

END OF THE STAGE 20W

Use population growth, feedback, capacity thresholds, model assumptions, scenario analysis, tradeoffs as an evidence chain rather than as isolated facts.

The Final Briefing

BEFORE THE STAGE — TAG

Catch Patel before he goes into the container lab

Patel, the molecular diagnostics lead, has the run sheet for last night's plates, which is the only record of which samples were re-tested and which were not. Once he is gowned and inside he is unreachable for three hours, and this morning's positives cannot be interpreted without knowing which of them are repeats.

PLAN CARD 99W

Day 140. The mayor is on air in an hour, and behind that broadcast sit five months of conclusions that are not equally solid. The route of transmission is settled. The animal source is not, and Castellanos has said so in writing. One thing said in public early — that this began with the hospital cluster — did not survive Oduya's sewer sample. This stage you go through the claims one at a time and decide what the city keeps funding after the emergency budget closes. Say too little and the next outbreak arrives with nobody watching for it.

BRIEFING 21W

The mayor, hospitals, and public must receive one integrated account of what happened, what remains uncertain, and which actions should continue.

WHAT YOU DECIDE 13W

Produce a claim-by-claim evidence package with explicit residual uncertainties and durable surveillance responsibilities.

15.1 Disposition the final claims

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A ROOM · ATTEST

SCENE 33W

The emergency is closing, and the final evidence wall contains four signed claims that will enter Riverton's permanent record. Two independent checks remain before publication. Their scientific evidence chains are not equally complete.

GUIDE 63W

Four signed claims are going into the permanent record and two checks remain. Open each claim and read what backs it now. Some rest on measurements from this outbreak, some on a chain of inference, and some on the strength of the wording alone. Hold what the evidence does not support, and check where a wrong claim would do the most lasting damage.

ASK 14W

Which claims need verification before the report can carry them at their proposed strength?

BEHIND THE BUTTON 134W

Why the permanent record raises the bar. These claims will be cited by people who cannot see the evidence behind them, in reviews and in the next outbreak's planning. A claim that is retracted later has already done its work, and the retraction rarely travels as far.

What proposed strength means. The same finding can be written as 'is', 'is consistent with' or 'cannot be excluded'. Verification decides which of those the evidence has earned, so a claim can pass by being weakened rather than by being checked.

Why two checks and four claims is the actual skill. Choosing is a judgement about consequence: which of these, if wrong, would misdirect the next response? That is what the last two slots are for, and spending them on the shakiest claim is not automatically right.

TAKES AS READ

a signed statement records a claim but does not independently verify the condition · strong conclusions need evidence chains that can survive one failed source

VERDICT 90W

A final report should not force every claim to the same confidence. The transmission route is backed by independent evidence and can support operational planning. A severe-risk subgroup from one small observational sample needs representative confirmation before it becomes citywide policy. A treatment benefit measured only through eight weeks cannot establish longer duration without follow-up. The suspected reservoir remains a live hypothesis and can stay in the report as unresolved. Verification capacity belongs on critical claims whose current evidence chain is weakest, not on claims that are already independently reconstructed.

TAKEAWAY 13W

Scientific claims should be communicated at the strength their independent evidence can support.

15.2 Build the public explanation

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON · SEQUENCE

SCENE 37W

The mayor goes live in an hour. The transmission route is settled, the animal reservoir is not, and treatment has only eight weeks of follow-up. Every sentence in the draft will be replayed after the emergency ends.

GUIDE 62W

All four things will be said, so the order is what is being decided. Ask of each how much weight its evidence can carry: three agreeing streams, one small sample, or a plausible hypothesis. Every sentence will be replayed after the emergency ends. Publishing the unconfirmed part with the confidence of the settled part is what makes the whole briefing unbelievable later.

ASK 11W

Order the briefing by how much weight each claim can carry.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

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TAKES AS READ

an audience judges a claim by how it was supported, once it knows

VERDICT 54W

The briefing is ordered by how much the evidence will carry, strongest first, because that is the order in which a listener can tell one claim from another. Publishing the reservoir with the confidence of the transmission route is what makes the whole document unbelievable when the reservoir turns out to be somewhere else.

TAKEAWAY 19W

A public account is ordered by the strength of its evidence, not by the order the work happened in.

15.3 Fund the post-crisis legacy

EPIDEMIOLOGY & EVOLUTION · EPIDEMIOLOGY OPERATIONS · A PERSON · CHOICE

SCENE 36W

The emergency budget ends in three weeks. Unprotected contracts and sampling programs stop with it. Sequencing, wastewater surveillance, field ecology and clinical readiness all want the last outbreak-specific line. The council votes before the month closes.

GUIDE 55W

Four capabilities, and three of them worked this time. Ask of each whether it would have worked on a different outbreak. Each stream sees a different part of transmission, and their value comes from being compared. Note also what the cheapest item on the list costs: records and methods are nearly free and disappear first.

ASK 11W

Which capability should the city protect when the emergency budget closes?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

capability that is not funded to continue stops when the emergency does

VERDICT 85W

The durable capability is an integrated surveillance system rather than one favored instrument. Clinical, wastewater and genomic streams see different parts of transmission. They become more useful when compared through shared metadata and analysis. Field ecology should remain available when a One Health signal appears, while routine clinical readiness belongs in the broader health system. The emergency line should protect cross-stream early warning and interpretation

capacity that would otherwise disappear. Records and methods should be preserved as a low-cost requirement rather than a competing project.

TAKEAWAY 25W

Legacy funding should preserve an integrated early-warning capability and the ability to interpret its signals, not just the instrument that was most visible this time.

15.4 Why did the assay turn positive?

MOLECULAR DIAGNOSTICS · A ROOM · CALLBACK · TRACE

SCENE 36W

Dr. Ravi Patel has patient wells, two controls and an independent platform on one screen. One ward has already cleared isolation rooms. If this run is wrong, patients will be moved before the morning team arrives.

GUIDE 60W

Open each result to see what its run depended on. Keep the ones that would still stand if one workflow step were contaminated, and untick the rest. Then name the dependency the failing ones share. A ward is already clearing isolation rooms, so both halves matter: a false positive moves patients, and discarding a true one leaves an outbreak unwatched.

ASK 12W

Which results share one workflow dependency, and what independent evidence survives it?

BEHIND THE BUTTON 137W

How a shared workflow makes results agree wrongly. Wells processed in the same batch pass through the same extraction, the same reagents and the same instrument. Contamination anywhere in that chain appears in every well downstream of it, and the pattern looks like a cluster of genuine positives.

What the controls are actually for. A negative control that stays negative rules out contamination affecting the whole batch. A positive control shows the chemistry worked. Neither can rule out a contamination event that entered after the controls were placed, which is why the independent platform is on the screen.

Why independence beats repetition. Running the same assay again on the same extract shares every dependency, so it will reproduce a contaminated result perfectly. A different platform on a different extract is the only evidence that survives the question.

TAKES AS READ

a no-template control contains no patient sample · two methods with different chemistry can fail for different reasons · DNA structure and replication — taken as read

VERDICT 86W

A no-template control has no patient material, so a target signal there must come from the workflow or reagents. That same path is shared by the patient amplification wells, which makes their positive signals ambiguous. The known positive control still shows that the chemistry can amplify target, but it cannot rescue a contaminated run. The independent platform uses different preparation and chemistry, so its negative result remains informative. The correct inference comes from tracing dependencies, not from counting how many wells on one plate are positive.

TAKEAWAY 13W

Agreement is not independent evidence when several results inherit the same contamination path.

END OF THE STAGE 19W

Use evidence synthesis, causal chains, uncertainty, ethics, communication, long-term monitoring as an evidence chain rather than as isolated facts.

THE CLOSING CARD

Forty-two days with no new case, and the outbreak was declared over on a Tuesday afternoon. The treatment works. The case definition held from the third week to the last. The animal reservoir was settled in the end — not by the briefing, but by the survey the briefing said was still needed.

The clinics stayed open, the surveillance stayed funded, and the curve that made the city frightened in March is now a figure in a report somebody else will learn from.

Card text only — no options, boards or answers. 12,538 words of card prose across 61 stops. Generated from themes/outbreak_riverton by tools/make-cardbook.mjs.